



KINCOPPAL-ROSE BAY
SCHOOL OF THE SACRED HEART

2010

**HIGHER SCHOOL CERTIFICATE
MID COURSE EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – $2\frac{1}{2}$ hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Start a new booklet for each question

Total marks – 96

- Attempt Questions 1 – 8
- All questions are of equal value

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Total Marks – 96

Attempt Questions 1-8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

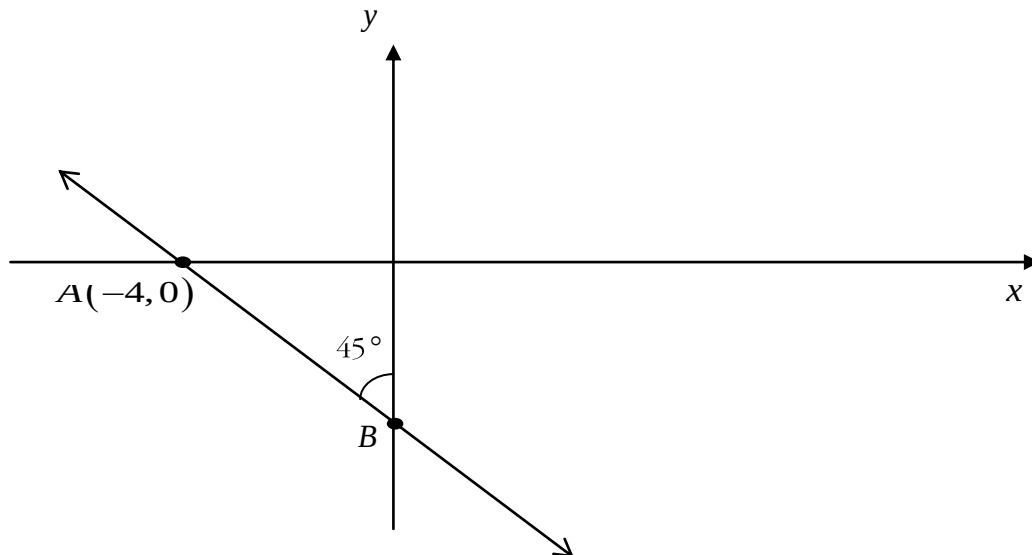
Question 1 (12 marks) Use a SEPARATE writing booklet	Marks
(a) Evaluate $\left(\frac{1}{\pi^{2.5}} - 1\right)^2$ correct to 3 significant figures.	2
(b) Express $\frac{\sqrt{3}}{\sqrt{7} - \sqrt{2}}$ with a rational denominator.	2
(c) Solve $ 3x - 7 \leq 4$	2
(d) State the domain of the function $y = \log_e x$	1
(e) If $\tan \theta = \frac{5}{12}$ and $\cos \theta > 0$, find the exact value of $\sin \theta$	2
(f) Solve $9^{2x-3} = 27^x$	3

End of Question 1

Question 2 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) Point A has coordinates $A(-4, 0)$. A line is drawn through point A as shown in the diagram below that intersects the y -axis at an angle of 45° .

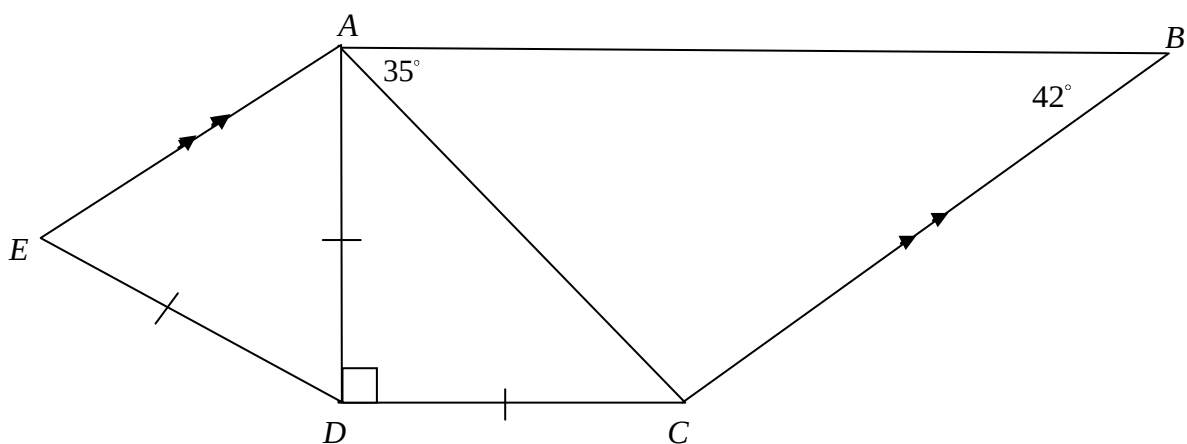


- | | | |
|--|---|---|
| (i) | Find the gradient of the line AB and state the coordinates of the point B .
You must show working out. | 2 |
| (ii) | Find the equation of the line that is parallel to AB passing through $(-4, -4)$ | 2 |
| If AB is the diameter of a circle, find: | | |
| (iii) | The coordinates of the centre of the circle | 1 |
| (iv) | The equation of the circle with diameter AB | 2 |
| (v) | Show that the line in part (ii) is a tangent to the circle with diameter AB . | 2 |

Question 2 continues on page 3

Question 2 (continued)

- (b) In the diagram below $ED = AD = DC$, $\angle ADC = 90^\circ$ and $AE \parallel BC$.
 $\angle ABC = 42^\circ$, $\angle CAB = 35^\circ$



Find the size of $\angle AED$. Give reasons for your answer.

3

End of Question 2

Question 3 (12 marks) Use a SEPARATE writing booklet

Marks

(a) Differentiate with respect to x

(i) $\frac{3}{x^5}$ **1**

(ii) $(5 - x^2)^7$ **2**

(ii) $x^3 e^{4x}$ **2**

(b) Use Simpson's Rule with five function values to approximate

$$\int_0^4 f(x) dx \text{ where } f(x) = \sqrt{x}$$

Give your answer correct to 1 decimal place. **3**

(c) Evaluate

$$\int_0^1 (e^{5x} - e^{-x}) dx$$

Give your answer in exact form. **2**

(d) Find

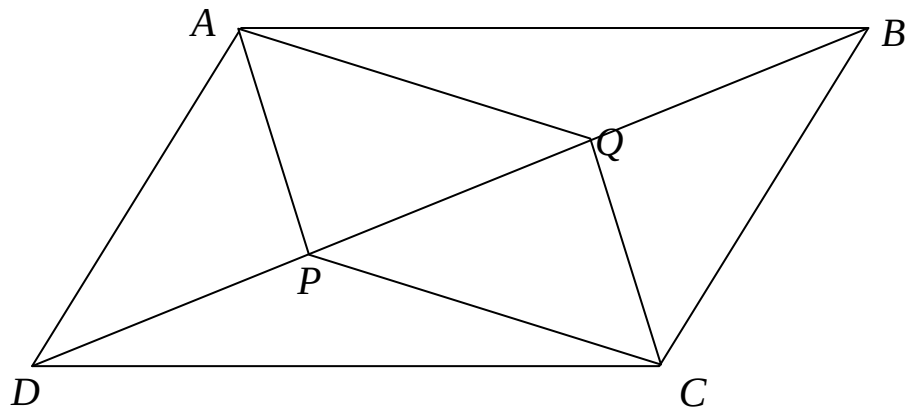
$$\int \frac{x^7 - x}{3x^5} dx$$
 2

End of Question 3

Question 4 (12 marks) Use a SEPARATE writing booklet

Marks

(a)



$ABCD$ is a parallelogram. Points P and Q lie on the diagonal BD such that $PD = QB$.

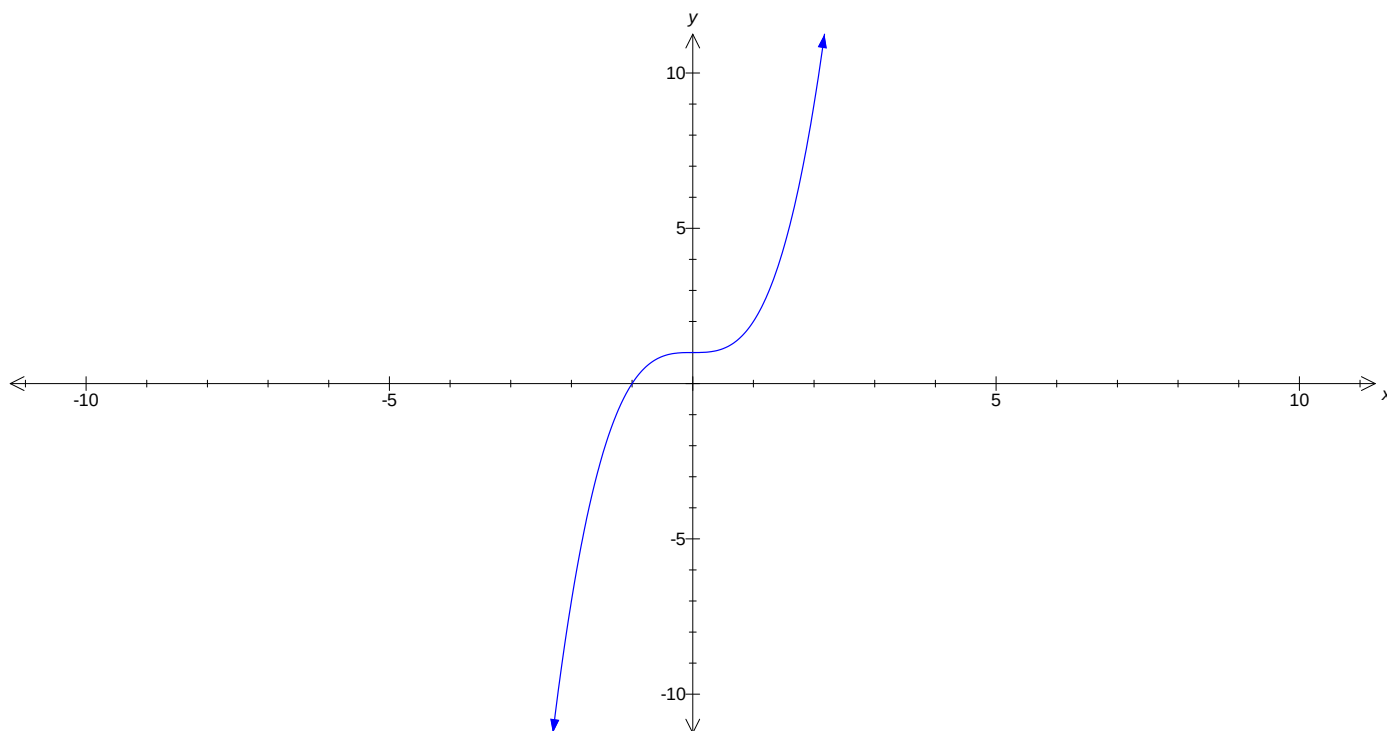
- | | | |
|-------|--|----------|
| (i) | Copy or trace the diagram into your answer booklet and indicate all the information given on your diagram. | 1 |
| (ii) | Prove that $\triangle ABQ \equiv \triangle CDP$ stating the congruency test used. | 3 |
| (iii) | Hence show that $AQ = CP$ | 1 |
| (b) | Find the values of k if the roots of the equation $2x^2 - 5x + k = 0$ are real and different. | 2 |
| (c) | Find the values of a, b and c for which $3x^2 + 5x - 10 \equiv ax(x + 3) + bx^2 + c(x + 1)$ | 2 |

Question 4 continues on page 6

Question 4 (continued)

- (d) (i) State the equation of the function shown below.

1



- (ii) Use the answer sheet page to accurately sketch on the same graph, $y = |x - 1|$

1

- (iii) Hence solve $x^3 \leq |x - 1| - 1$

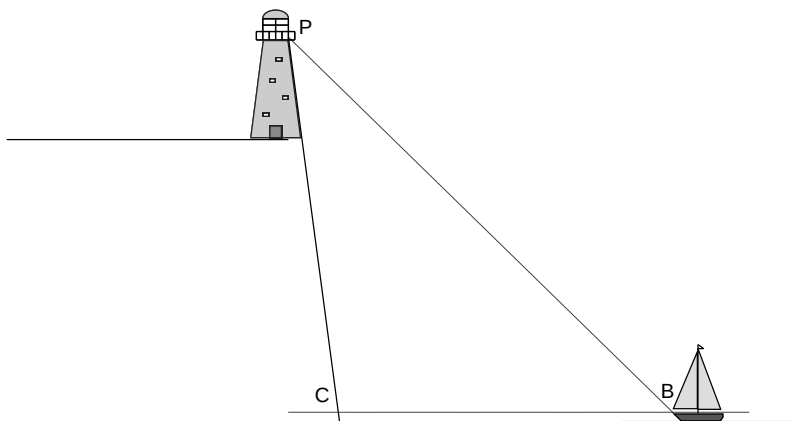
1

End of Question 4

Question 5 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) A lighthouse sits at the top of a cliff which slopes at 105° from the horizontal. From the bow of the ship (B) the angle of elevation of the platform of the lighthouse (P) is 46° . Using radio signals the ship determines the distance PB is 800 metres.



- (i) Copy the diagram and mark all of the relevant information above on it. 1
- (ii) Calculate the distance BC of the bow from the base of the cliff
Give your answer correct to the nearest metre. 2
- (b) Find the equation of the normal to the curve $y = e^{5x-1}$ at the point $x = 1$ 3
- (c) Show that $\frac{(1 + \tan^2 \theta) \cot \theta}{\operatorname{cosec}^2 \theta} = \tan \theta$ 3
- (d) Solve $3\sin^2 \theta + 2\sin \theta = 1$ for $0 \leq \theta \leq 360^\circ$
Give your answers in degrees and minutes. 3

End of Question 5

Question 6 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) If α, β are the roots of the equation $2x^2 - 14x - 1 = 0$, find the value of
- | | | |
|-------|--------------------------------------|----------|
| (i) | $\alpha\beta$ | 1 |
| (ii) | $\frac{1}{\alpha} + \frac{1}{\beta}$ | 2 |
| (iii) | $\alpha^2 + \beta^2$ | 1 |
- (b) Given the quadratic polynomial $(x - 4)^2 = -8(y + 1)$, find the coordinates of the:
- | | | |
|-------|--|----------|
| (i) | Vertex | 1 |
| (ii) | Focus (hint draw a diagram) | 2 |
| (iii) | Find the coordinates of the end points of the latus rectum | 2 |
- (c) The area bounded by the parabola $y = 1 - x^2$, and the x and y axes in the first quadrant is rotated about the x -axis. Find the exact volume of the solid formed. **3**

End of Question 6

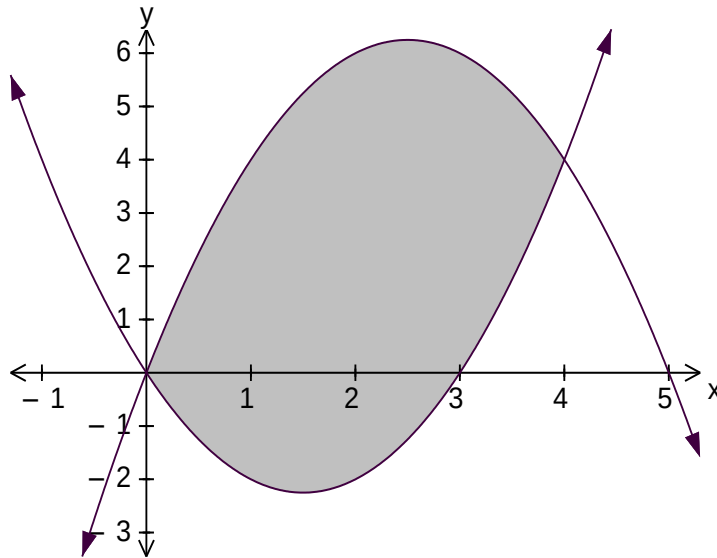
Question 7 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) If $\log_a 2 + 2\log_a x - \log_a 6 = \log_a 3$ find the value of x .

2

(b)



- (i) The curves $y = x^2 - 3x$ and $y = 5x - x^2$ intersect at the point $(0, 0)$.
Show that they also intersect at the point $(4, 4)$.

1

- (ii) Find the exact shaded area enclosed between the two curves.

3

- (c) A closed cylindrical can is made from a sheet of metal with an area of $600\pi \text{ cm}^2$.
Given that the surface area of a closed cylinder is $SA = 2\pi r^2 + 2\pi rh$ and the volume is $V = \pi r^2 h$:

- (i) Show that the volume of the can is given by
 $V = 300\pi r - \pi r^3$

2

- (ii) Find the exact maximum volume of the cylinder.

3

- (iii) Justify your answer to part (ii).

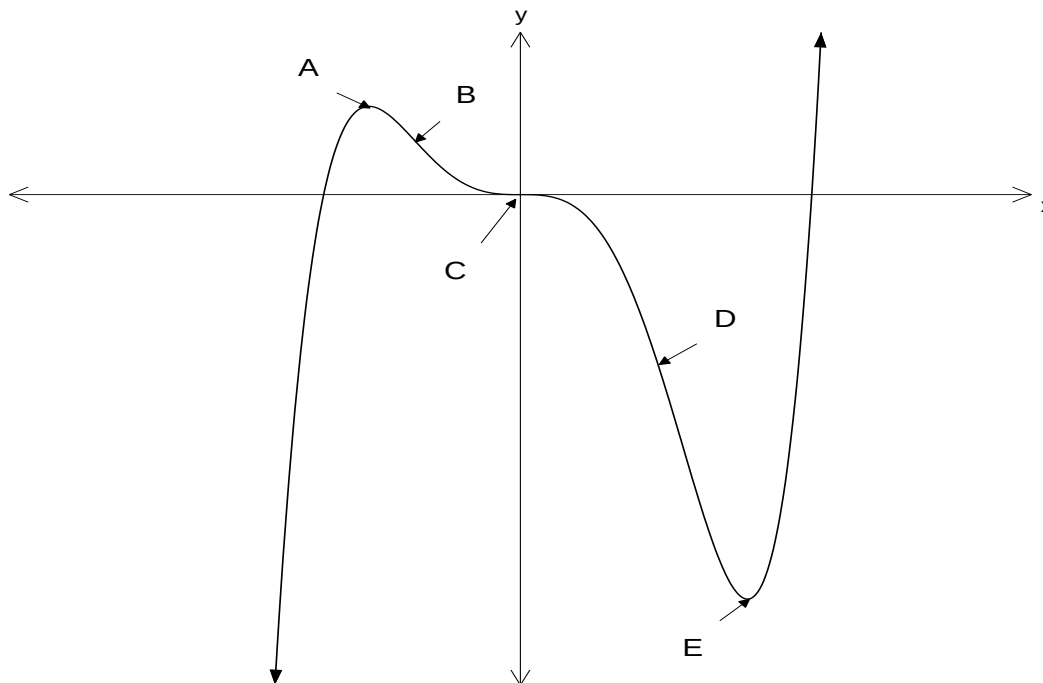
1

End of Question 7

Question 8 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) The graph of the curve $y = f(x)$ is drawn below.



- | | | |
|---|---|----------|
| (i) | Name the points of inflexion. | 1 |
| (ii) | When is the graph decreasing? | 1 |
| (iii) | Sketch the gradient function. | 2 |
|
(b) Consider the curve $y = x^3 - 6x^2 + 8$ | | |
| (i) | Find any stationary points and determine their nature. | 3 |
| (ii) | Show that there is one point of inflexion and find its co-ordinates. | 2 |
| (iii) | Hence sketch the curve for the domain $-1 \leq x \leq 5$ indicating important features. | 2 |
| (iv) | Find the maximum value of the function in the domain $-1 \leq x \leq 5$. | 1 |

End of Examination



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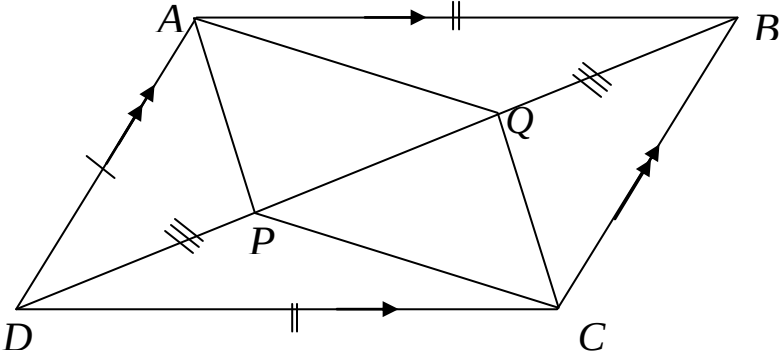
HIGHER SCHOOL CERTIFICATE
HALF YEARLY EXAMINATION

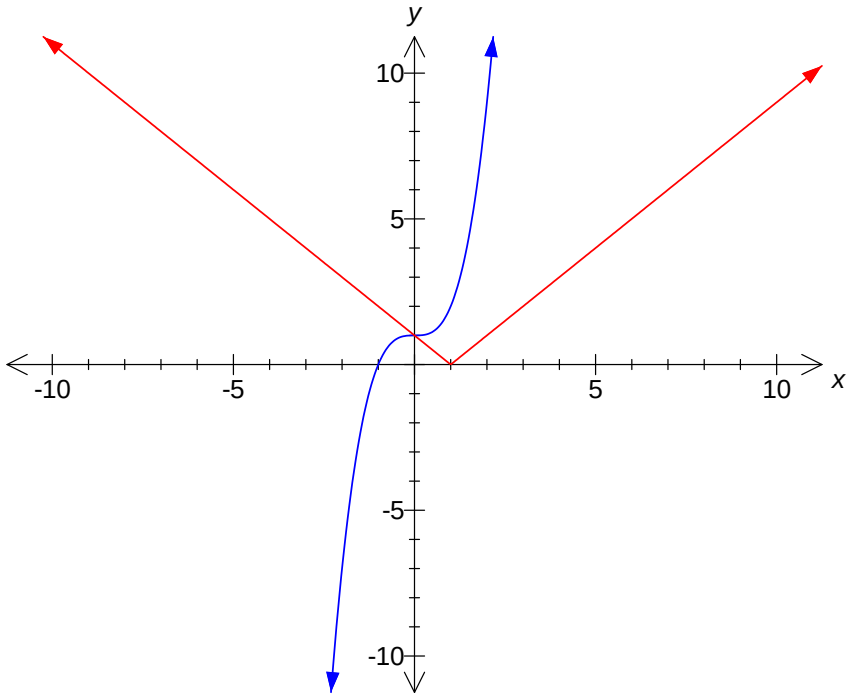
Mathematics SOLUTIONS

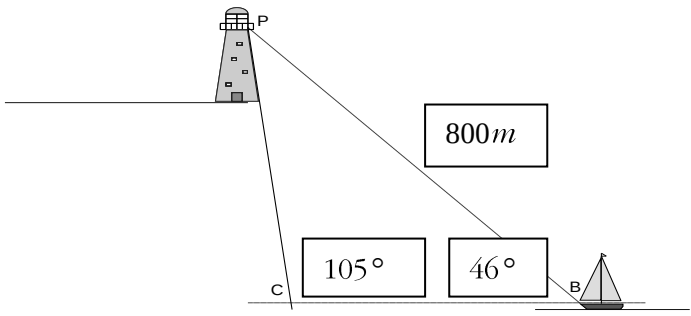
Question 1	Criteria	Marks
1(a)	0.88893905 $= 0.889$	1 1
1(b)	$\frac{\sqrt{3}[\sqrt{7} + \sqrt{2}]}{7 - 2}$ $= \frac{\sqrt{21} + \sqrt{6}}{5}$	1 1
1(c)	$x \leq \frac{11}{3} = 3\frac{2}{3} \text{ and } x \geq 1$ or $1 \leq x \leq \frac{11}{3}$	1 1
1(d)	Domain : $x > 0$, for all real x	1
1(e)	$H = \sqrt{5^2 + 12^2} = 13$ $\therefore \sin \theta = \frac{5}{13}$	1 1
1(f)	$3^{2(2x-3)} = 3^{3x}$ $4x - 6 = 3x$ $x = 6$	1 1 1

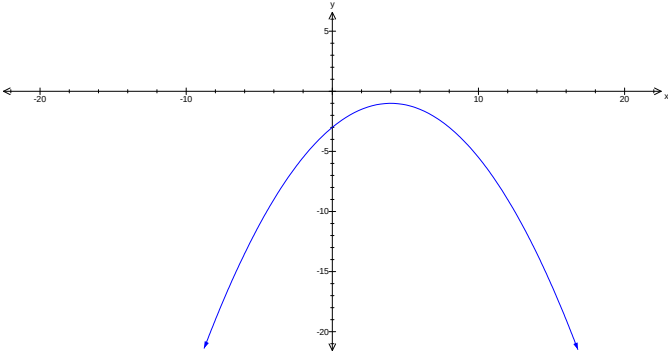
Question 2	Criteria	Marks
2(a)(i)	$m = \tan 135^\circ = -1$	1
2(a)(ii)	$y + 4 = -1(x + 4)$ $y = -x - 8$ or $x + y + 8 = 0$	1 1
2(a)(iii)	$A(-4, 0)$ $B(0, -4)$ midpoint $AB = (-2, -2)$	1
2(a)(iv)	distance centre to $A = \sqrt{(-2 + 4)^2 + (-2 + 0)^2}$ $= \sqrt{8}$ or $2\sqrt{2}$ Circle: $(x + 2)^2 + (y + 2)^2 = 8$	1 1
2(a)(v)	$\frac{ x + y + 8 }{\sqrt{1^2 + 1^2}}$ to $(-2, -2) = \frac{ -2 - 2 + 8 }{\sqrt{2}}$ $= \frac{4}{\sqrt{2}}$ or $2\sqrt{2}$ Since the distance from the line to the centre of the circle is equal to the radius, the line must be a tangent to the circle	1 1
2(b)	$\angle EAB = 180^\circ - 42^\circ$ (cointerior angles $\angle EAB$ and $\angle ABC$; $EA \parallel CB$) $= 138^\circ$ $\angle DAC = 45^\circ$ (equal angles of isosceles $\triangle ADC$; angle sum $\triangle ADC$) $\angle DAE = 138^\circ - (45^\circ + 35^\circ)$ (sum of adjacent angles) $= 58^\circ$ $\angle AED = 58^\circ$ (equal angles of isosceles $\triangle ADE$)	1 1 1

Question 3	Criteria	Marks															
3(a)(i)	$-15x^{-6} = \frac{-15}{x^6}$	1															
3(a)(ii)	$7(5-x^2)^6 \times -2x$ $= -14x(5-x^2)^6$	1 1															
3(a)(iii)	$3x^2e^{4x} + 4x^3e^{4x}$ $= x^2e^{4x}(3+4x)$	1 1															
3(b)	<table border="1"><tr><td>a= 0</td><td>1</td><td>2</td><td>3</td><td>b=4</td></tr><tr><td>0</td><td>1</td><td>$\sqrt{2}$</td><td>$\sqrt{3}$</td><td>2</td></tr><tr><td>y_0</td><td>y_1</td><td>y_2</td><td>y_3</td><td>y_4</td></tr></table> $n = 4, h = \frac{4-0}{4} = 1$ $\int_0^4 f(x) dx = \frac{1}{3} \left[(0+2) + 4 \left[1 + \sqrt{3} \right] + 2\sqrt{2} \right]$ $= 5.2522.....$ $= 5.3$	a= 0	1	2	3	b=4	0	1	$\sqrt{2}$	$\sqrt{3}$	2	y_0	y_1	y_2	y_3	y_4	1 1 1
a= 0	1	2	3	b=4													
0	1	$\sqrt{2}$	$\sqrt{3}$	2													
y_0	y_1	y_2	y_3	y_4													
3(c)	$\left[\frac{1}{5} e^{5x} + e^{-x} \right]_0^1$ $= \left(\frac{1}{5} e^5 + e^{-1} \right) - \left(\frac{1}{5} e^0 + e^0 \right)$ $= \frac{1}{5} e^5 + e^{-1} - 1 \frac{1}{5}$	1 1															
3(d)	$\frac{1}{3} \int (x^2 - x^{-4}) dx$ $= \frac{1}{3} \left[\frac{x^3}{3} + \frac{x^{-3}}{3} \right] + C$ $= \frac{1}{9} \left[x^3 + \frac{1}{x^3} \right] + C$ $= \frac{1}{9} x^3 + \frac{1}{9x^3} + C$	1 1															

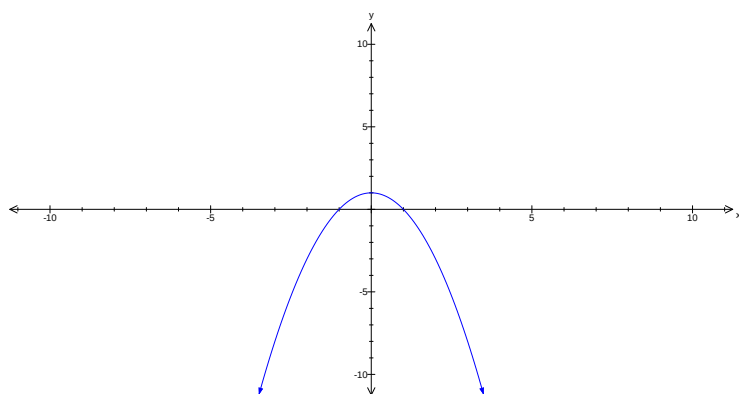
Question 4	Criteria	Marks
4(a)(i)		1
4(a)(ii)	$DC = AB$ (opposite sides of parallelogram $ABCD$ are equal) $DP = QB$ (given) $\angle PDC = \angle ABQ$ (alternate $\angle$'s are equal; $AB \parallel DC$) $\therefore \triangle ABQ \equiv \triangle CDP$ (two sides and the included angle equal SAS)	1 1 1
4(a)(iii)	$AQ = CP$ (corresponding sides of congruent triangles are equal)	1
4(b)	$\Delta = (-5)^2 - 4 \times 2 \times k = 25 - 8k$ for real and different roots $\Delta > 0$ $\therefore 25 - 8k > 0$ $k < \frac{25}{8}$ or $k < 3\frac{1}{8}$	1 1
4(c)	$3x^2 + 5x - 10 \equiv ax^2 + 3ax + bx^2 + cx + c$ $\equiv x^2(a+b) + x(3a+c) + c$ $c = -10$ $3a + c = 5 \Rightarrow a = 5$ $a + b = 3 \Rightarrow b = -2$ $a = 5, b = -2, c = -10$	1 1

4(d)(i)	$y = x^3 + 1$	1
4(d)(ii)		
4(d)(iii)	$x^3 + 1 \leq x - 1 $ from the graph the cubic is below the absolute value for: $x \leq 0$	1

Question 5	Criteria	Marks
5(a)(i)		1
5(a)(ii)	$\angle CBP = 29^\circ$ $\frac{BC}{\sin 29^\circ} = \frac{800}{\sin 105^\circ}$ $BC = 401.52948 \dots \text{metres}$ $BC = 402 \text{metres}$	1 1
5(b)	$m = \frac{dy}{dx} = 5e^{5x-1}; x=1, m_1 = 5e^4$ $\therefore m_2 = \frac{-1}{5e^4}$ $x=1, y = e^4$ $\therefore y - e^4 = \frac{-1}{5e^4} [x - 1]$ $y = \frac{-1}{5e^4} [x - 1] + e^4 = \frac{1}{5e^4} + e^4 - \frac{x}{5e^4}$	1 1 1
5(c)	$\frac{(\sec^2 \theta) \frac{1}{\tan \theta}}{\sin^2 \theta} = \frac{\left(\frac{1}{\cos^2 \theta} \right) \times \frac{\cos \theta}{\sin \theta}}{\sin^2 \theta}$ $= \left(\frac{\sin^2 \theta}{\cos^2 \theta} \right) \times \frac{\cos \theta}{\sin \theta} = \frac{\sin \theta}{\cos \theta} = \tan \theta$	1, 1 1
5(d)	$3\sin^2 \theta + 2\sin \theta - 1 = 0$ $(3\sin \theta - 1)(\sin \theta + 1) = 0$ $\sin \theta = \frac{1}{3}, \sin \theta = -1$ $\theta = 19^\circ 28', 160^\circ 32', 270^\circ 00'$	1 1 1

Question 6	Criteria	Marks
6(a)(i)	$\alpha\beta = -\frac{1}{2}$	1
6(a)(ii)	$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta}$ $= \frac{7}{-\frac{1}{2}} = -14$	1 1
6(a)(iii)	$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ $= 7^2 - 2\left(-\frac{1}{2}\right) = 50$	1
6(b)(i)	$(4, -1)$	1
6(b)(ii)	$a = 2; (4, -3)$ 	2
6(b)(iii)	length of latus rectum = 8; 4 left of focus and 4 right of focus $(8, -3)$ and $(0, -3)$	2

6(c)



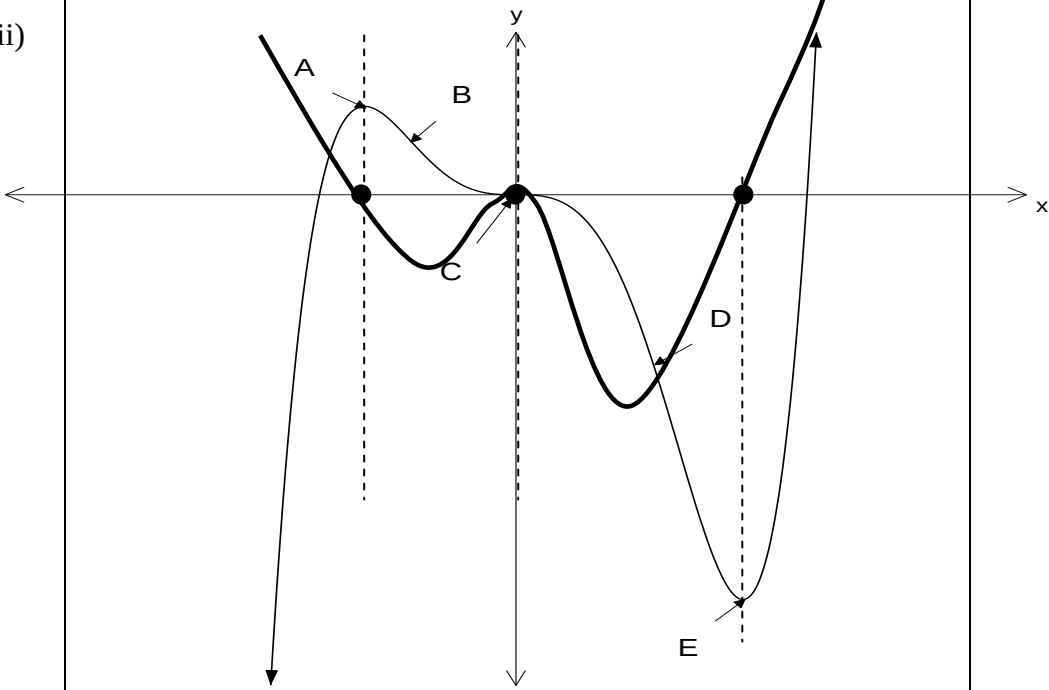
$$\begin{aligned}
 V &= \pi \int_0^1 (1-x^2)^2 dx \\
 &= \pi \int_0^1 1 - 2x^2 + x^4 dx \\
 &= \pi \left[x - \frac{2}{3}x^3 + \frac{1}{5}x^5 \right]_0^1 \\
 &= \pi \left[1 - \frac{2}{3} + \frac{1}{5} - 0 \right] \\
 &= \frac{8}{15} \pi
 \end{aligned}$$

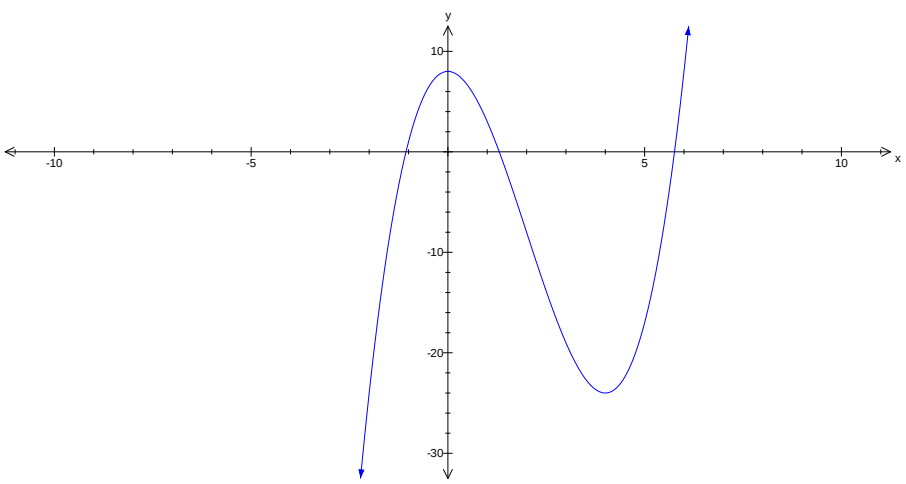
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1

1

Question 7	Criteria	Marks
7(a)	$\log_a \left[\frac{2x^2}{6} \right] = \log_a 3$ $\frac{x^2}{3} = 3$ $x^2 = 9$ $x = \pm 3$ $x \neq -3$ $\therefore x = 3$	1 1
7(b)(i)	$4^2 - 3(4) = 4, 5(4) - 4^2 = 4$	1
7(b)(ii)	$A = \int_0^4 (5x - x^2) - (x^2 - 3x) dx$ $= \int_0^4 (8x - 2x^2) dx = \left[4x^2 - \frac{2x^3}{3} \right]_0^4$ $= \left\{ 4(4)^2 - \frac{2(4)^3}{3} \right\} - \left\{ 4(0)^2 - \frac{2(0)}{3} \right\}$ $= 64 - \frac{2(64)}{3} - 0 = 21\frac{1}{3} \text{ units}^2$	1 1
7(c)(i)	$600\pi = 2\pi r^2 + 2\pi rh$ $h = \frac{300 - r^2}{r}$ $V = \pi r^2 \left[\frac{300 - r^2}{r} \right] = 300\pi r - \pi r^3$	1 1
7(c)(ii)	$\frac{dy}{dx} = 300\pi - 3\pi r^2 = 0$ $r^2 = 100, \therefore r = 10\text{cm}, \text{ reject } r = -10\text{cm}$ $\therefore V = 3000\pi - 1000\pi = 2000\pi \text{cm}^3$	1 1
7(c)(ii)	$\frac{d^2y}{dx^2} = -6\pi r = -6\pi(10) = -60\pi < 0$ So V is a maximum.	1

Question 8	Criteria	Marks
8(a)(i)	<i>B, C, D</i>	1
8(a)(ii)	Between <i>A</i> and <i>C</i> and <i>C</i> and <i>E</i>	1
8(a)(iii)	 1 correct zeros 1 correct shape	2
8(b)(i)	$\frac{dy}{dx} = 3x^2 - 12x$ $\frac{dy}{dx} = 0 \Rightarrow 3x(x - 4) = 0$ $\therefore x = 0, 4 \quad y = 8, -24$ $\frac{d^2y}{dx^2} = 6x - 12$ $\text{at } x = 0 \quad \frac{d^2y}{dx^2} < 0 \quad \Rightarrow (0, 8) \text{ is max}$ $\text{at } x = 4 \quad \frac{d^2y}{dx^2} > 0 \quad \Rightarrow (4, -24) \text{ is min}$	1 1 1

8(b)(ii)	$\frac{d^2y}{dx^2} = 0$ $6x - 12 = 0 \Rightarrow x = 2$ Test: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td></tr><tr><td>$\frac{d^2y}{dx^2}$</td><td><0</td><td>0</td><td>>0</td></tr></table> There is a change in concavity and hence $(2, -8)$ is a point of inflexion	x	1	2	3	$\frac{d^2y}{dx^2}$	<0	0	>0	1 1
x	1	2	3							
$\frac{d^2y}{dx^2}$	<0	0	>0							
8(b)(iii)	 1 for endpoints: $(-1, 1)$ and $(5, -17)$ 1 for sketch showing important features turning point, y-intercept	1 1								
8(b)(iv)	Max value in the domain is 8	1								